Task 2 Report

Team: T26

Dept: Cs

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Introduction

This project implements the Back-Propagation learning algorithm on a multi-layer neural network (MLP) that classifies penguins into one of three species: Adelie, Chinstrap, and Gentoo. The dataset contains 50 samples per species, and each sample has five features: CulmenLength, CulmenDepth, FlipperLength, BodyMass, and OriginLocation.

For each species, the first 30 non repeated samples are used to train the network and the remaining 20 samples are used to test it, which gives 90 training samples and 60 testing samples in total. The performance of the classifier is evaluated using the confusion matrix and the overall accuracy on the test samples.

Submitted files

- **code.py** - data preparation, the MLP class (forward, backward, training, prediction), and the confusion matrix.
- **gui.py** - the graphical user interface that uses code.py.

Main Observations after modeling

There are some interesting observations when applying both algorithms on different parameter combinations.

Best Accuracy for Each Activation Function

Best accuracy obtained with each activation function (training: first 30 samples per class, testing: remaining 20 samples per class). Accuracy values are in %.

Activation Function	Train Accuracy	Test Accuracy	LR	Epochs	#Layers	#HiddenNodes
Sigmoid	97.78	98.33	0.05	300	2	5, 5
Tanh	100.00	96.67	0.1	100	1	5

Bias: Yes (Sigmoid), Yes (Tanh).

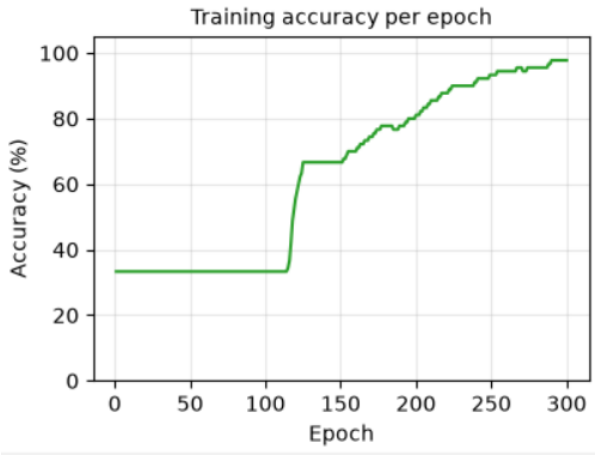
How the values were found: A grid search tested 256 combinations of activation functions, bias settings, network architectures, learning rates, and epochs. Models were ranked by test accuracy, with ties broken by training accuracy, fewer epochs, and smaller network size. Several Tanh configurations achieved 96.67%, while two Sigmoid configurations tied for the best result.

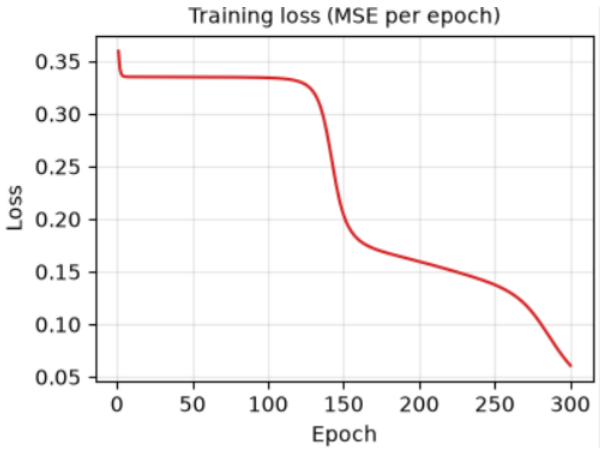
See full 256 combinations' results in the data/tuning_results.csv file

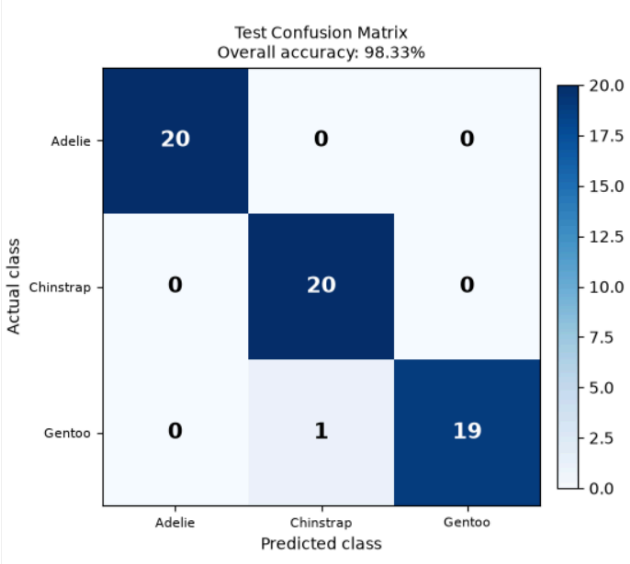
Detailed Results

Sigmoid

The best result with the Sigmoid function was obtained with learning rate 0.05, 300 epochs, 2 hidden layers with 5, 5 neurons each, and bias on. It reached 97.78% training accuracy and 98.33% testing accuracy.

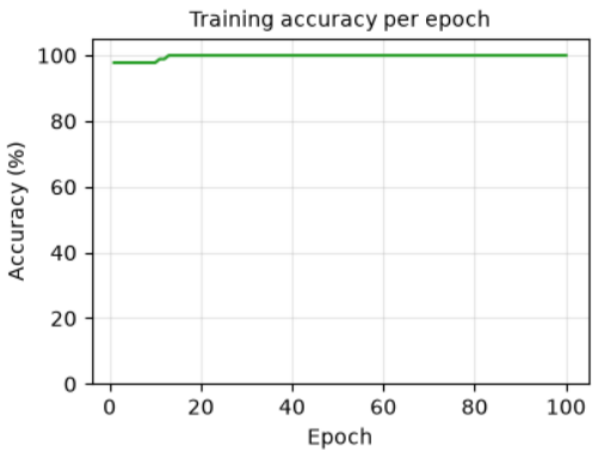


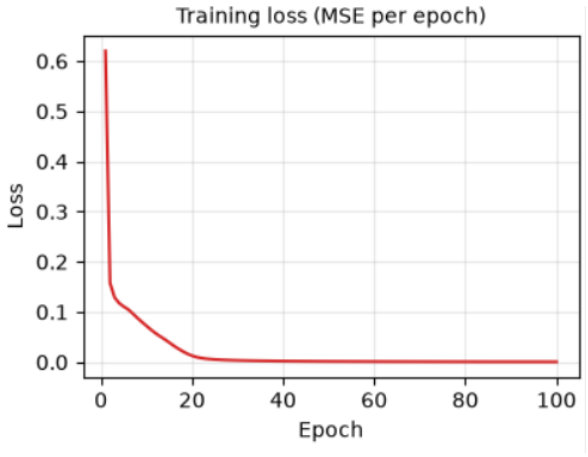


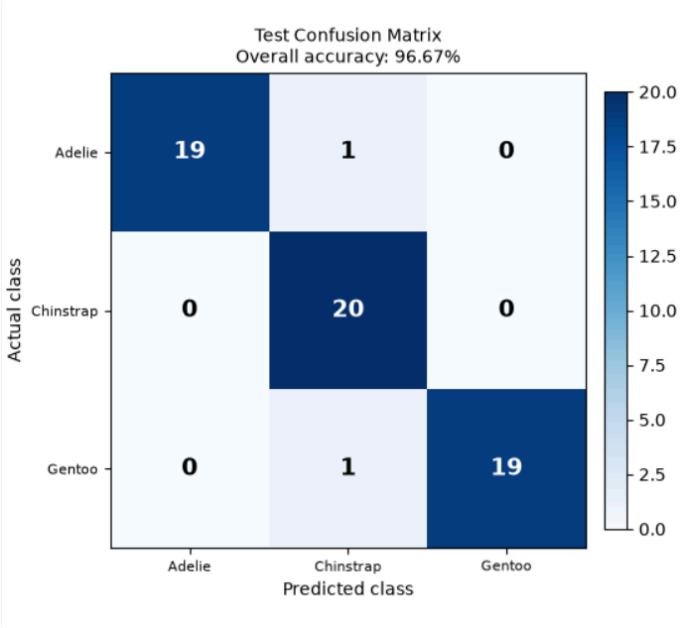


Hyperbolic Tangent (Tanh)

The best result with the Tanh function was obtained with learning rate 0.1, 100 epochs, 1 hidden layer with 5 neurons, and bias on. It reached 100.00% training accuracy and 96.67% testing accuracy.

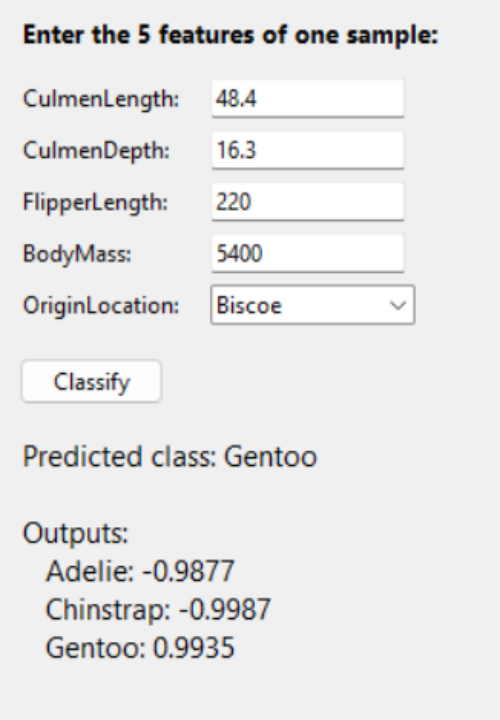






Classifying a single sample

Example of classifying one sample: after training the best Sigmoid network, the sample CulmenLength = 48.4, CulmenDepth = 16.3, FlipperLength = 220, BodyMass = 5400, OriginLocation = Biscoe was entered in the Classify a sample tab. The network predicted Gentoo, which is the correct species. The output values were -0.9877 (Adelie), -0.9987 (Chinstrap) and 0.9935 (Gentoo), so the Gentoo neuron had the maximum output.



Conclusion & Limitations

The best result was obtained with the Sigmoid function: 98.33% test accuracy and 97.78% training accuracy, using two hidden layers of 5 neurons each, a learning rate of 0.05, 300 epochs and bias enabled. The best Tanh network reached 96.67% test accuracy and 100% training accuracy with one hidden layer of 5 neurons, a learning rate of 0.1, 100 epochs and bias enabled. The gap between the two is only one test sample, so both functions solve the task well. The species are well separated by the standardized culmen, flipper and body-mass features, which explains why almost every reasonable setting reached more than 90%. Tanh trained faster, was less sensitive to the learning rate, and gave a simpler network, while Sigmoid gave the single highest test accuracy. Bias should be kept enabled, especially with Tanh.

Limitations

- Each result comes from a single 30/20 split with 60 test samples, so one misclassified sample changes the accuracy by about 1.7%.
- The best parameters were selected using the test accuracy itself, so the reported best accuracy can be slightly optimistic.
- OriginLocation was encoded as ordered integers, which creates an artificial order between the islands.
- The initial weights use a fixed random seed, so each result depends on one particular initialization.